Physics 225, Spring 2026: Homework 12 - Solutions
Due 4/30/26

All solutions must clearly show the steps or reasoning you used to arrive at your result. You will lose points for poorly written solutions or incorrect reasoning/explanations; answers given without explanation will not be graded.

Name:
Section:

1. **Index notation practice.** Here you will use the Levi-Civita symbol to quickly derive several useful derivative identities: this is a good example of the power of index notation!

   (a) Derive the identity $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ for any vector field $\mathbf{A}$. The index heights work out most easily if you treat $\mathbf{A}$ to have lower indices A_k, which is equivalent to thinking of $\mathbf{A}$ as a row vector. This should be a *very* quick computation, no more than a few lines! The curl in index notation is $(\nabla \times \mathbf{A})^i = \epsilon^{ijk}\partial_j A_k$, and the gradient is ∂_i. So

   $$\nabla \cdot (\nabla \times \mathbf{A}) = \partial_i \epsilon^{ijk}\partial_j A_k = \epsilon^{ijk}\partial_i \partial_j A_k.$$

   Note that this is exactly the same form as the expression from lecture for the curl of a gradient, once we move the ∂_i past the Levi-Civita symbol, which we are allowed to do because the components of ϵ^{ijk} are constant. So the same arguments from lecture hold: the last expression has every $\partial_i\partial_j$ term compensated by a $-\partial_j\partial_i$, and thus it vanishes.

   (b) There is a lowered-index version of the Levi-Civita symbol, ϵ_{ijk}, defined identically to ϵ^{ijk}. Show that the product of the two Levi-Civita symbols can be written in terms of the Kronecker delta:

   $$\epsilon_{abc}\epsilon^{ijk} = \delta^i{}_a\delta^j{}_b\delta^k{}_c + \delta^i{}_b\delta^j{}_c\delta^k{}_a + \delta^i{}_c\delta^j{}_a\delta^k{}_b - \delta^i{}_b\delta^j{}_a\delta^k{}_c - \delta^i{}_a\delta^j{}_c\delta^k{}_b - \delta^i{}_c\delta^j{}_b\delta^k{}_a \tag{1}$$

   Hint: the only values the left-hand side can take are -1, 1, or 0. Every term on the right-hand side corresponds to a unique permutation of (a,b,c), so you just have to check the signs and verify that the right-hand side vanishes whenever the left-hand side does.
   Note that there are $3! = 6$ permutations of (a, b, c), each of which corresponds to a term on the right-hand side. The even permutations have sign $+1$, and the odd permutations have sign -1.
   To see how this works out, consider plugging in indices and evaluating $\epsilon_{123}\epsilon^{213}$. Every term on the right-hand side vanishes except for $-\delta^i{}_b\delta^j{}_a\delta^k{}_c$, because that term matches the upper indices with appropriate lower indices. Since $i = b$, $j = a$, and $k = c$ for our choice of indices, the Kronecker deltas all evaluate to 1, and the right-hand side is -1. Likewise, the left-hand side is -1 since $\epsilon^{123} = 1$ is an even permutation, and $\epsilon^{213} = -1$ is an odd permutation.
   In general, for any choice of indices (i, j, k) that is a permutation of $(1, 2, 3)$, a choice of (a, b, c) that is the same sign as the permutation of (i, j, k) gives $(\pm 1)^2 = 1$ on the left-hand side, and corresponds to exactly one term on the right-hand side with a positive coefficient. If (a, b, c) has the opposite sign permutation, the left-hand side is $(1)(-1) = -1$ and matches exactly one term on the right-hand side with negative coefficient.

Now suppose (i,j,k) is *not* a permutation of $(1,2,3)$: for example, $(1,1,2)$. The Levi-Civita symbol vanishes in that case, $\epsilon^{112}=0$, so the left-hand side is 0. On the right-hand side, a Kronecker delta can set $i=a$ and $j=b$, but it can also set $i=b$ and $j=a$. These two types of terms appear with opposite signs, so they cancel and the right-hand side also vanishes.

(c) By contracting the first index of Eq. (1), derive the contraction identity

$$\epsilon_{ibc}\epsilon^{ijk} = \delta^j{}_b\delta^k{}_c - \delta^j{}_c\delta^k{}_b. \tag{2}$$

Hint: you will have some traces in your expression. Remember your result from problem 2b of HW 3!

After all the hard work in part (b), we now just have to contract both sides, which we can do by simply plugging in $a=i$:

$$\epsilon_{abc}\epsilon^{ijk} = \delta^i{}_i\delta^j{}_b\delta^k{}_c + \delta^i{}_b\delta^j{}_c\delta^k{}_i + \delta^i{}_c\delta^j{}_i\delta^k{}_b - \delta^i{}_b\delta^j{}_i\delta^k{}_c - \delta^i{}_i\delta^j{}_c\delta^k{}_b - \delta^i{}_c\delta^j{}_b\delta^k{}_i.$$

Let's look at the structure of each term. The first term and the fifth term both have $\delta^i_i = 3$ (this was the trace of the identity from HW 3). The other terms each have a contraction of two Kronecker deltas, which are just instructions to set the uncontracted indices equal: for example, in the second term, $\delta^i{}_b\delta^k{}_i = \delta^k{}_b$. Doing this contraction on each term yields

$$\epsilon_{ibc}\epsilon^{ijk} = 3\delta^j{}_b\delta^k{}_c + \delta^j{}_c\delta^k{}_b + \delta^j{}_c\delta^k{}_b - \delta^j{}_b\delta^k{}_c - 3\delta^j{}_c\delta^k{}_b - \delta^j{}_b\delta^k{}_c$$

Terms 1, 4, and 6 are $\delta^j{}_b\delta^k{}_c$ with a coefficient of $(3-1-1)=1$, and likewise terms 2, 3, and 5 are $\delta^j{}_c\delta^k{}_b$ with a coefficient of $1+1-3=-1$. Adding these up yields exactly the identity we are meant to prove.

(d) The *Laplacian operator* ∇^2 is defined to act on vectors as

$$\nabla^2\mathbf{F} \equiv \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\mathbf{F} \Longleftrightarrow \partial^i\partial_i F_a. \tag{3}$$

Use your identity in Eq. (2) to derive the "curl-of-a-curl" identity

$$\nabla\times(\nabla\times\mathbf{F}) = \nabla(\nabla\cdot\mathbf{F}) - \nabla^2\mathbf{F}. \tag{4}$$

We will use this equation in an important way over the next 2 weeks!

First write the left-hand side in index notation, using both versions the Levi-Civita symbol to take the curl:

$$(\nabla\times(\nabla\times\mathbf{F}))_b = \epsilon_{bci}\partial^c(\nabla\times\mathbf{F})^i = \epsilon_{bci}\partial^c(\epsilon^{ijk}\partial_j F_k)$$

Note that the final expression has only one free index b, which matches the the free index on the original expression. To us the identity from part (c), we reorder the indices on the first Levi-Civita symbol: $\epsilon_{ibc} = \epsilon_{bci}$, since to get from bci to ibc we have to permute i to the right twice, and that keeps the sign of the permutation the same. We can then pass the second Levi-Civita symbol through the derivative, yielding $\epsilon_{ibc}\epsilon^{ijk}\partial^c\partial_j F_k$. Now we apply the identity and contract the resulting Kronecker deltas:

$$\epsilon_{ibc}\epsilon^{ijk}\partial^c\partial_j F_k = \left(\delta^j{}_b\delta^k{}_c - \delta^j{}_c\delta^k{}_b\right)\partial^c\partial_j F_k = \partial^c\partial_b F_c - \partial^c\partial_c F_b$$

The order of partial derivatives in the first term can be exchanged to give $\partial^c\partial_b F_c = \partial_b\partial^c F_c = [\nabla(\nabla\cdot\mathbf{F})]_b$, and the second term is exactly how we've defined the Laplacian, $\partial^c\partial_c F_b = (\nabla^2\mathbf{F})_b$.

2. **Line integral practice.** (30 points) Compute the line integrals $\int_C \mathbf{F}\cdot d\mathbf{r}$ of the following vector fields $\mathbf{F}$ over the corresponding paths C:

(a) $\mathbf{F} = \mathbf{r}$, along the straight line C defined by $x(t) = t$, $y(t) = 3t-1$, $z(t) = 2t$, from the points $(1,2,2)$ to $(3,8,6)$.

We are given the parametrization of the path C in terms of t, so we now just need to compute $d\mathbf{r}(t)/dt = (1,3,2)$. The vector field $\mathbf{F}$ evaluated along C is $\mathbf{F} = (x(t), y(t), z(t)) = (t, 3t-1, 2t)$. The initial point corresponds to $t = 1$ and the final p oint t o $t = 3$. The line integral is

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_1^3 \mathbf{F}(\mathbf{r}(t))\cdot\frac{d\mathbf{r}}{dt}\,dt = \int_1^3 (t(1) + (3t-1)(3) + (2t)(2))\,dt = \int_1^3 (14t-3)\,dt = 50.$$

(b) $\mathbf{F} = (2xy, z^2, 3)$, along the path C which is the intersection of the paraboloid $z = x^2+y^2$ with the plane $y = 2x$, from $(0,0,0)$ to $(2,4,20)$.

A useful parametrization of the path $\mathbf{s}(t)$ is $t = x$, so that $y = 2t$ and $z = t^2 + (2t)^2 = 5t^2$. The derivative along the path is $d\mathbf{s}(t)/dt = (1,2,10t)$ and the path goes from $t = 0$ to $t = 2$. The vector field evaluated along the path is $\mathbf{F} = (2(t)(2t), (5t^2)^2, 3) = (4t^2, 25t^4, 3)$. The line integral is

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_0^2 \mathbf{F}(\mathbf{s}(t))\cdot\frac{d\mathbf{s}}{dt}\,dt = \int_0^2 \left(4t^2(1) + 25t^4(2) + 3(10t)\right)dt = \frac{1172}{3}.$$

(c) $\mathbf{F} = (-y, x, 0)$ counterclockwise along the circle C defined by $x^2+y^2 = 4$ from $(2,0)$ to $(-2,0)$. *Hint: choose a good coordinate system!*

Since we have a circular path of radius 2 which stays in the xy-plane, we should use polar coordinates with $\rho = 2$ and $t = \phi$, and take $\mathbf{s}(t) = (2\cos t, 2\sin t)$. The derivative is $d\mathbf{s}(t)/dt = (-2\sin t, 2\cos t)$ and the path goes from $t = 0$ to $t = \pi$. The vector field evaluated along the path is $\mathbf{F} = (-2\sin t, 2\cos t)$, which is in fact the same as the derivative. So the line integral is

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_0^\pi \left((-2\sin t)^2 + (2\cos t)^2\right)dt = 4\int_0^\pi dt = 4\pi.$$